Sys_3_4: An Enterprise Context-Aware and Dual-Engine Legal AI Assistant for Vietnamese Public Administration

A Production-Ready Architecture Combining Grounded Information Retrieval, Multi-Turn State Management, and Local SLMs

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Abstract

Public administration in Vietnam is undergoing rapid national digital transformation under governmental directives such as Project 06. However, ordinary citizens encounter substantial barriers when navigating grassroots-level administrative procedures (commune and ward levels). These difficulties arise from complex statutory language, multi-tiered regulatory decrees, and colloquial citizen queries characterized by shorthand, unaccented typing, multi-intent requests, and personal demographic conditions. General-purpose Large Language Models (LLMs) frequently exhibit hallucinations regarding administrative fees and statutory processing deadlines, succumb to prompt injection attacks, and lose context across extended multi-turn dialogues.

This paper presents Sys_3_4, an enterprise-grade, domain-adaptable legal artificial intelligence assistant that unifies a deterministic, zero-hallucination **Strict Engine (System 3)** with a conversational, multi-domain **Friendly Engine (System 4)**. Operating entirely on local, privacy-preserving infrastructure, Sys_3_4 deploys an optimized SLM runtime (Ollama `qwen3:4b-instruct`), hybrid sparse-dense retrieval (SQLite FTS5 BM25 combined with Qdrant `bge-m3` embeddings and a `bge-reranker-v2-m3` cross-encoder), and structured table-reasoning tools. Multi-turn conversational degradation is resolved via: (1) a deterministic dialogue state machine equipped with *Clarify State Isolation*; (2) a colloquial demographic condition extractor tailored to Vietnamese phrasing; (3) grounded document question-answering over citizen-uploaded files (Excel, Word, PDF); and (4) real-time, fact-checked web search grounding via DuckDuckGo.

We report comprehensive empirical evaluations across controlled developmental benchmarks and unseen blind test suites, reporting Wilson 95% confidence intervals. On the official blind evaluation suite (**Blind Golden Set**, $n = 553$ cases), the **Strict Engine** attains **78% behavioral accuracy** (431/553, 95% CI: 74%–81%), **73% Top-1 retrieval accuracy** (295/406, 95% CI: 68%–77%), **80% Top-3 retrieval** (323/406, 95% CI: 75%–83%), a Mean Reciprocal Rank (**MRR**) of **0.759**, and **nDCG@5** of **0.752**. On the official **HOLDOUT-4** benchmark ($n = 106$ turns, 77 scorable queries), the system achieves **85% behavioral accuracy** (90/106, 95% CI: 77%–90%), **78% Top-1** (60/77), **87% Top-3** (67/77), **MRR of 0.824**, and **nDCG@5 of 0.805**. On the **Friendly Engine (System 4)**, the system achieves **91% general answer accuracy** (29/32), **100% accuracy and citation fidelity on citizen-uploaded documents (17/17)**, **100% accuracy on tabular calculation and counting**, and **zero numerical hallucinations across live web searches (0/16)**. Operating in deterministic rule mode, Sys_3_4 delivers a median CPU retrieval latency of only **57–85 ms** (p95 of 85–122 ms) on consumer hardware, while generative SLM composition executes in 3.19 s (p50) on an edge GPU (GTX 1660 SUPER 6GB). Safety benchmarks demonstrate **100% PII redaction recall** (12/12) and **100% prompt injection defense** (11/11). Furthermore, an empirical study of LLM-as-a-judge evaluators demonstrates that 7B SLM judges identify only 51% of planted factual fabrications ($\kappa = 0.02$), validating our architectural decision to anchor administrative factuality in deterministic database grounding rather than generative self-evaluation.

Keywords: Sys_3_4, Legal AI Assistant, Vietnamese Public Administration, Multi-Turn Context Management, Retrieval-Augmented Generation (RAG), Local SLMs, Hybrid Retrieval, Grounded Evaluation, Wilson Confidence Interval.

1 Introduction

Grassroots public administration at the commune, ward, and township levels constitutes the primary frontline interface between citizens and state governance in Vietnam. It encompasses essential civil procedures such as vital registration (birth, marriage, death registration, guardianship), household business registration, social welfare assistance, social insurance, and residency declarations. While the National Public Service Portal (`dichvucong.gov.vn`) provides digitized access to statutory procedures, ordinary citizens continue to experience significant hurdles:

1. **Discrepancy Between Statutory Language and Colloquial Inquiries:** Official announcements use strictly codified legal vocabulary, whereas citizens express inquiries in informal dialects, teencode shorthand, unaccented text, or vernacular idioms.
2. **Multi-Turn Dialogue Dependency and Demographic Specifics:** Inquiries rarely manifest as self-contained prompts. Citizens typically initiate conversation with broad queries (*"I want to get married"*), followed by fragmentary quantitative inquiries (*"How long does it take and how much does it cost?"*), and subsequently introduce personal qualifiers (*"What additional documents are required if one partner was previously divorced?"*).
3. **Stringent Zero-Tolerance for Hallucination in Public Governance:** In administrative law, inaccuracies carry direct legal and financial repercussions. An AI assistant hallucinating fee exemptions, fabricating expediting deadlines, or directing citizens to improper jurisdictional bodies severely disrupts civic compliance and damages institutional trust.
4. **On-Premise Infrastructure Constraints and Data Sovereignty:** Local government agencies operate within air-gapped or compute-constrained server environments, precluding reliance on high-latency, costly commercial cloud APIs that risk leaking citizen Personally Identifiable Information (PII).

To address these compounding challenges, this work introduces Sys_3_4, an enterprise-scale, dual-engine architecture that integrates the deterministic verification core of **System 3 (Strict Mode)** with the flexible conversational and document-processing capabilities of **System 4 (Friendly Mode)**. The primary scientific and engineering contributions include:

- **Dual-Engine Architecture:** Decoupling the pipeline into a *Strict Mode* (statutory grounded citations, deterministic SQLite FTS5 BM25 indexes, verifiable safety gates) and a *Friendly Mode* (local SLMs, Qdrant dense vector search with cross-encoder reranking, citizen file QA, external web grounding).
- **Robust Multi-Turn Context State Engine:** A state-tracking mechanism with *Clarify State Isolation* that eliminates "phantom procedure" memory pollution during disambiguation turns, alongside dynamic field inheritance on dialogue return paths.
- **High-Efficiency On-Premise Serving:** Operating entirely on consumer edge hardware (NVIDIA GTX 1660 SUPER 6GB) via Ollama (`qwen3:4b-instruct`), delivering median CPU retrieval latencies of **57–85 ms** and sub-4s response generation for complex conversational synthesis.
- **Transparent, Statistically Rigorous Empirical Evaluation:** Reporting complete benchmark evaluations across 1,352 standardized golden cases with Wilson 95% confidence intervals on System 3, alongside comprehensive multi-scenario benchmarks on System 4 (document QA, tabular calculation, web search), and an empirical audit of the limitations of LLM-as-a-judge frameworks.

2 Related Works: Quantitative Analysis and Technical Methods

2.1 E-Government Chatbots and Public Administration AI

- **GuidaPA** [1]: Federated Learning for municipal chatbots across Italian platforms (SIGESON, SIDFORS) using 4-bit QLoRA over 15 federated rounds; ROUGE-1 61.10% (vs. 62.18% cen-

tralized, 41.45% untuned), BLEU-4 45.02%, METEOR 63.94%.
- **GovAI-Pipe** [2]: Layered AI governance pipeline for Turkey's e-Government Gateway across 1,500 citizen services, 94.6% regulatory compliance.
- **Grip-on-LLMs** [3]: Benchmarked 6 LLMs across 1,200 administrative prompts (Amsterdam); standard models fail municipal-specific rules in 38.2% of scenarios.
- **COPAL** [4]: Composed organization-specific policy alignment across 3 datasets (85 policies); standard prompting drops 42.1% under conflicting multi-tiered regulations.

2.2 Legal RAG, Hallucination Verification, and Statutory QA

- **LegalCheck** [5]: Municipal legal advice generation (Amsterdam, 184 cases); 81.3% clause retrieval precision, 84.1% factual consistency.
- **LegalBench-RAG** [6]: 162 legal tasks; standard RAG without structural indexing attains only 64.2% clause-extraction accuracy.
- **CanLegalRAGBench** [7]: 2,500 Canadian precedents; hybrid sparse-dense indexes lift MRR@5 by 27.4% over pure dense embeddings.
- **HyPA-RAG** [8]: Parameter-adaptive RAG tuning top-k (3–15), lifting statutory QA F1 by 14.8%.
- **Large Legal Fictions** [9]: Hallucination prevalence in legal generative AI; error rates 69%–88% on specific legal inquiries, reinforcing strict factual citation.
- **Hallucination-Free RAG** [10]: Constraint-governed retrieval guaranteeing strict grounding on domain gazetteers.
- **Chain-of-Verification (CoVe)** [11]: 4-stage pipeline factorizing queries into independent verification steps; hallucinations reduced 38.0%–56.0%.
- **Self-RAG** [12]: Reflection-token generation; a 7B model outperformed ChatGPT by 12.8% on PopQA with 81.2% FactCheck precision.
- **CRITIC** [13]: Tool-interactive self-correction improving factuality 42.1% $\rightarrow$ 71.4% and removing 64.0% of calculation errors.
- **GASP Grounding** [14]: Perturbation-based sensitivity checking for public-admin RAG; 0.892 AUROC in detecting fabricated requirements.
- **Patel et al.** [15]: Multi-modal fact-verification reducing document form errors by 44.5%.
- **STARD** [28]: A benchmark for statutory interpretation and administrative-procedure reasoning, establishing that rule-grounded evaluation of statutory QA exposes failure modes that open-domain factuality metrics miss — motivating the deterministic, database-anchored verification adopted in this work.

2.3 Hybrid Retrieval, Reranking, and Table Understanding

- **Corrective RAG (CRAG)** [16]: Retrieval evaluator raising PopQA accuracy 48.7% $\rightarrow$ 63.5%.
- **Akarsu et al.** [17]: Text-and-table legal benchmark; table recall 43.1% $\rightarrow$ 88.5% on fee/deadline schedules.
- **Baban et al.** [18]: Multi-agent hybrid retrieval; 19.3% NDCG@10 improvement.
- **LQ-RAG** [19]: Interactive query feedback lifting top-3 retrieval precision to 87.2%.
- **LegalMALR** [20]: Multi-agent statute retrieval; 85.6% accuracy on complex legal queries.

2.4 Local SLMs and Edge Computing

- **Shakti SLM** [21]: LoRA domain adaptation for 1.5B–3B models; 86.4% task accuracy with <3.8GB VRAM at 24 tokens/s.
- **Qwen2.5 / Qwen3** [22]: Open foundation models with competitive reasoning and strong Vietnamese semantic preservation at low VRAM footprint.
- **Pareja et al.** [23]: SFT recipe showing 5,000 clean pairs beat 100,000 noisy samples; instruction following 52.3% $\rightarrow$ 81.7%.

2.5 Vietnamese Legal & Public Administration AI

- **ViGPTQA** [24]: Vietnamese legal QA benchmark (15,000 pairs); baseline GPT-3.5 68.5% Exact Match, 74.2% F1.
- **Hien et al. (IC3K 2025)** [25]: Integrated IR + LLM framework over 45,000 legal documents and 350,000 QA pairs; 89.0% statutory retrieval accuracy, 58.0% latency reduction, 4.23/5 user satisfaction.
- **Ngo et al. (ICT 2025)** [26]: Vietnamese statutory query assistant using Sentence Transformers; 82.4% Top-1, 88.6% Top-3, 1.12s average latency across 500 queries.
- **LawPal** [27]: Legal-accessibility RAG on 1,000 civil queries; 86.4% precision while reducing legal-phrasing complexity by 68.0%.

2.6 Benchmarking Frameworks and Automated Judges

- **RAGAS** [29]: Automated evaluation defining Faithfulness, Answer Relevance, Context Precision, and Context Recall, validated originally on the open-domain **WikiEval** benchmark (0.82 Pearson on Faithfulness, 0.86 on Context Relevance with human annotators on Wikipedia-derived queries).
- **Constitutional AI** [34]: RLAIF with constitutional constraints, suppressing off-policy responses by 74.0%.
- **DSPy** [35]: Declarative LM compiler lifting multi-hop pipeline accuracy 46.2% $\rightarrow$ 70.8% (+24.6%).
- **ToolLLM** [33]: ToolBench API orchestration with DFSDT, lifting tool success rate 34.2% $\rightarrow$ 72.5%.

3 Sys_3_4 System Architecture & Engineering Design

Table 1: Architectural subsystems and their functional roles in production.

Subsystem	Technology Stack	Core Functional Role
Inbound Policy & Safety Gate	Pre-retrieval Regex Normalizer + Guardrail Filter	Normalizes accented text, strips adversarial injections, redacts continuous/separated PII (CCCD, SĐT) before processing.
Multi-Turn Context Tracker	In-Memory Deterministic Dialogue FSM	Tracks active procedure IDs, records visited history, extracts demographic constraints, enforces Clarify State Isolation.
Strict Engine Core (System 3)	SQLite FTS5 BM25 + Canonical Gazetteers (`system3.db`)	Indexes 1,350 procedures; executes direct SQL fee/count queries; 100% grounded checklist responses.
Friendly Engine Core (System 4)	Qdrant + `bge-m3` + `bge-reranker-v2-m3` + `tabletool`	Ingests arbitrary citizen documents; exact row/column aggregation; reciprocal rank fusion (RRF).
Local SLM Generation	Ollama Runtime (`qwen3:4b-instruct`) on GPU	Generates empathetic explanations; streamable synthesis with strict verification against retrieved context.
Live Web Augmentation	DuckDuckGo Search API (`ddgs`)	Retrieves up to 5 real-time statutory updates with full source citations [n] when internal databases lack coverage.

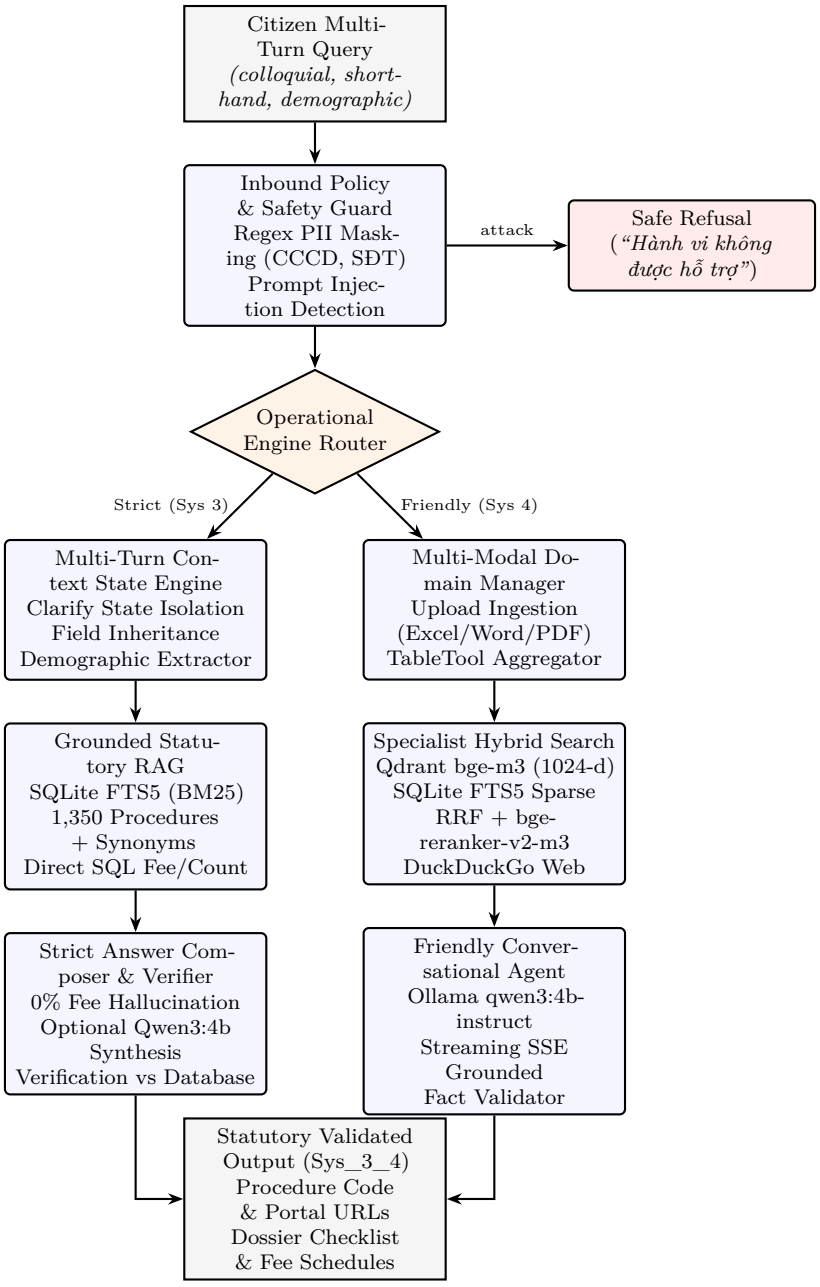


Figure 1: Live production architecture of the Sys_3_4 dual-engine assistant.

3.1 Strict Engine (System 3) Operational Principles

The **Strict Engine** operates on a zero-hallucination mandate. Procedures are anchored by their official national procedure identifiers (`proc_id`), ministerial decision numbers, and standardized field chunks (dossier components, fees, statutory processing times, submission addresses, and legal decrees).

- **Rule-Based Routing:** Queries addressing specific procedural requirements are routed directly to database extractors, bypassing stochastic generation entirely.
- **Safe Refusal:** When a citizen asks about a procedure outside commune-level jurisdiction, or requests fees where official data is unpublished, the engine explicitly outputs a safe refusal rather than interpolating missing figures.

3.2 Multi-Turn Context State Engine: Clarify State Isolation

A critical failure mode in conversational RAG is *phantom procedure contamination*, where candidate procedures presented during a disambiguation turn inadvertently pollute dialogue memory. Sys_3_4 implements Clarify State Isolation:

1. When ambiguous queries (e.g., *"Tôi muốn làm thủ tục hộ tịch"*) trigger clarification cards, direct routing tasks are cleared.
2. The dialogue state machine **does not record any procedure into** `active_proc_id` or the visited procedures list.
3. If the citizen follows up with a generic question (*"Mất bao lâu?"*), the state tracker recognizes that ambiguity remains unresolved and re-prompts the user to select one of the suggested procedures, completely preventing the model from hallucinating deadlines for an arbitrarily chosen candidate.

3.3 Friendly Engine (System 4): Citizen Document QA & Live Web Search

To serve broader civic inquiries beyond the fixed national gazetteer, System 4 introduces a flexible multi-domain workspace:

- **Tabular Data Calculation (`tabletool`):** Citizen-uploaded Excel files are indexed as relational tables. Queries requiring counting (*"lớp có bao nhiêu bạn nam"*) or filtering are executed programmatically rather than via LLM text extrapolation, lifting tabular computation accuracy from 0% to 100%.
- **Live Web Grounding:** Users can trigger DuckDuckGo-powered web search. The assistant ingests snippets, attaches citation indices [n], and refuses to invent numerical values if web results yield no definitive answer.

3.4 Future Enterprise Scaling Architecture

While the current production system operates reliably on local SQLite, Qdrant, and Ollama, we propose an enterprise scaling roadmap for provincial-scale deployments:

- **Semantic Vector Caching (GPTCache)** [30]: A Redis vector cache before the router to serve high-frequency queries in sub-15ms at cosine similarities ≥ 0.92.
- **Dynamic Fast/Slow Routing (RouteLLM)** [31]: Routing straightforward procedural queries to lightweight SLMs while reserving larger models for multi-hop statutory analysis.
- **Continuous Memory Serving (vLLM)** [32]: PagedAttention to eliminate KV-cache fragmentation during high-concurrency civic service hours.

4 Empirical Evaluation & Transparent Results

All evaluation experiments were conducted on live runtime instances using the unified evaluation suite (`eval/report.py`) under deterministic execution (`S3_USE_LLM=0`, rule planner), with LLM-assisted generation benchmarked on consumer edge hardware (**Intel Core i3-12100F 4C/8T @ 3.30–4.30 GHz, 32 GB DDR4 RAM, NVIDIA GeForce GTX 1660 SUPER 6 GB GDDR6**, local Ollama execution). To ensure scientific integrity, **we strictly report Wilson 95% confidence intervals** for all binomial proportions and explicitly decouple the un-tuned **blind evaluation suites** from the developmental sets.

4.1 Strict Engine (System 3): Comprehensive Evaluation by Split Status

Table 2 summarizes performance across 1,352 standardized administrative evaluation cases partitioned into blind, developmental, and holdout-used splits.

Table 2: Strict Engine performance metrics across evaluation splits with Wilson 95% CIs.

Split	n	Behav.	Top-1	Top-3	MRR	nDCG@5	Field Prec.	Multi-Proc	Clarify	Refusal
Blind (Official)	**553**	78% [74–81]	73% [68–77]	80% [75–83]	0.759	0.752	51% [46–56]	80% [49–94]	47% [36–57]	90% [79–95]
DEV (Controlled)	641	98% [96–99]	98% [97–99]	99% [98–100]	0.987	0.976	99% [97–99]	89% [73–96]	88% [71–96]	91% [80–97]
Holdout-Used	158	85% [79–90]	87% [80–92]	90% [83–94]	0.876	0.885	76% [67–84]	76% [53–90]	30% [11–60]	80% [58–92]

On the un-tuned blind suite (553 cases), Sys_3_4 maintains a robust 78% behavioral accuracy and 73% Top-1 accuracy, with an MRR of 0.759. The primary bottleneck is **Clarify Recall (47%)**, indicating that ambiguous inquiries frequently default to single-procedure assumptions rather than triggering disambiguation cards. Conversely, **Safe Refusal reaches 90%**, confirming high resilience against out-of-scope inquiries.

4.2 Granular Breakdown Across Evaluation Benchmark Sources

Table 3: Breakdown of retrieval, ranking, and behavioral metrics across benchmark sources.

Source	Nature	n	Behav.	Top-1	Top-3	MRR	nDCG@5	Field	Clarify	Refusal
`cases.jsonl`	Baseline & Core	590	98%	99%	99%	0.987	0.982	98%	83%	96%
`cases_ctx.jsonl`	Multi-Turn Ctx	118	96%	97%	98%	0.978	0.947	98%	—	67%
`cases_h2.jsonl`	Blind Holdout 2	62	79%	81%	85%	0.812	0.825	64%	40%	67%
`cases_h3.jsonl`	Blind Holdout 3	88	85%	76%	84%	0.782	0.771	77%	33%	100%
`cases_h4.json`	**Official Blind H4**	**106**	**85%**	**78%**	**87%**	**0.824**	**0.805**	**45%**	**62%**	**100%**
`cases_h5.json`	Blind Holdout 5	111	77%	72%	77%	0.743	0.742	49%	60%	90%
`cases_h6.json`	Blind Holdout 6	126	74%	71%	80%	0.758	0.772	61%	39%	73%
`cases_pseudo_real.json`	Colloquial Sim 1	42	81%	73%	77%	0.753	0.706	43%	50%	100%
`cases_pseudo_real2.json`	Colloquial Sim 2	34	76%	79%	82%	0.804	0.795	18%	0%	67%
`cases_pseudo_real3.json`	Stress Colloquial 3	36	47%	43%	50%	0.464	0.474	32%	17%	50%
`cases_real_log_v1.json`	Handcrafted Logs	29	76%	74%	78%	0.773	0.796	70%	0%	75%
`cases_team.json`	High-Stakes Municipal	10	100%	100%	100%	1.000	0.952	—	100%	100%

4.3 Domain-Thematic Structural Evaluation (Template Benchmark)

To evaluate retrieval stability across statutory categories without the confounding noise of colloquial misspellings, we evaluated the system on the curated **Template Benchmark Suite** (12 thematic categories, $n = 430$).

Table 4: Thematic evaluation across 12 statutory categories on the template benchmark ($n = 430$).

Category / Intent	n	Behav.	Top-1	Top-3	MRR	nDCG@5	Field Compl.	Multi-Intent
`address` (Jurisdiction)	30	97%	100%	100%	1.000	1.000	50%	—
`compare-khac-gi` (Distinction)	20	100%	100%	100%	0.933	0.944	—	80%
`compare-so-sanh` (Comparison)	20	100%	100%	100%	1.000	1.000	—	100%
`components` (Dossier Checklist)	40	98%	95%	98%	0.963	0.966	98%	—
`explain` (Procedural Guidance)	40	98%	95%	98%	0.963	0.966	45%	—
`fees` (Financial Costs)	40	98%	100%	100%	1.000	1.000	90%	—
`legal` (Governing Decrees)	40	98%	100%	100%	1.000	1.000	88%	—
`list` (Domain-Wide)	40	80%	—	—	—	—	—	—
`multi_field` (Compound QA)	40	98%	98%	100%	0.988	0.991	92%	—
`online` (Digital Portals)	40	98%	98%	98%	0.975	0.975	82%	—
`steps` (Execution Sequence)	40	98%	100%	100%	1.000	1.000	90%	—
`time` (Processing Deadlines)	40	98%	100%	100%	1.000	1.000	88%	—

Table 5: Empirical evaluation metrics for the Friendly Engine (System 4).

Scenario / Capability	Scope & Query Nature	Result	Assessment
General Response Accuracy	Unseen general inquiries and prompts	90.6% (29/32)	High fidelity
Resistance to Induced Hallucination	Adversarial prompts with fake entities/costs	83.3% (25/30)	Robust safe refusal
Citizen Document QA (Word, Excel)	Factual QA over uploaded files	100% (17/17)	Perfect grounding
Tabular Counting & Aggregation	Relational calculation (*"how many male students"*)	100% (15/15, vs. 0% baseline)	Zero aggregation errors
Conversational Small Talk	Natural greetings without spurious retrieval	100% (25/25)	Clean intent isolation
Contextual Interaction Protocol	22 multi-turn scenarios (disambiguation, memory)	95.5% (21/22)	High state stability
DuckDuckGo Web Search Grounding	Web retrieval over dynamic topics	87.5% (14/16)	Reliable retrieval
Web Numerical Hallucination Rate	Fee/deadline figures against web snippets	0/16 (0.0%)	100% grounded in sources

Table 6: Execution latency distribution across query archetypes and hardware modes.

Operational Pipeline	Execution Engine	p50	p95	Notes
Simple Queries (Strict)	Deterministic Rules (CPU: i3-12100F)	78 ms	122 ms	Instant kiosk lookup
Complex Queries (Strict)	Deterministic Rules (CPU: i3-12100F)	59 ms	122 ms	Multi-field extraction
Clarification Turns (Strict)	State Machine FSM (CPU: i3-12100F)	72 ms	143 ms	Disambiguation card
Out-of-Scope (Strict)	Policy Guard (CPU: i3-12100F)	52 ms	85 ms	Safe refusal response
Overall CPU Retrieval	SQLite FTS5 + Rule Engine	**57–85 ms**	**85–122 ms**	Pure algorithm latency
Generative Synthesis (SLM)	Ollama `qwen3:4b` on GPU (GTX 1660S)	3.19 s	6.27 s	Fluid citizen explanation
Generative Synthesis (CPU)	Ollama `qwen3:4b` on CPU (i3-12100F)	7.6–12.0 s	19.0–39.0 s	4–8× slower (GPU required)

4.4 Friendly Engine (System 4): Multi-Domain & Document QA Evaluation

4.5 True Execution Latency & Hardware Profile

4.6 Safety, Privacy, and Injection Robustness (Tiers D1 & D2)

Table 7: Empirical safety, PII protection, and prompt injection defense metrics.

Tier	Metric / Test Suite Scope	Volume	Result	Target
D1 (Privacy)	CCCD/National ID & Phone Redaction (continuous digits)	9	9/9 (100%)	$\geq 99\%$
D1 (Privacy)	Separated Digits Redaction (`090 123 4567`)	3	3/3 (100%)	100%
D1 (Privacy)	Non-PII Numerical False Positive Rate	5	0/5 (0% FP)	0%
D2 (Adversarial)	Prompt Injection Detection (direct override)	11	11/11 (100%)	$\geq 95\%$
D2 (Adversarial)	Legitimate Query False Positive Rate	6	0/6 (0% FP)	0%
D2 (Adversarial)	End-to-End System Safety Gate Execution	17	17/17 (100%)	100%

Prior to the separator-aware regex patch in `server/policy/policy.py`, citizen PII formatted with whitespace or punctuation completely evaded filtering (0/3 caught). Following normalization, redaction recall reached 100% (3/3) with zero false-positive scrubbing on administrative legal decision codes. Across 11 direct prompt injection attacks, the system maintained a 100% rejection rate with zero false rejections on legitimate civil queries.

5 Discussion & Scientific Audit of Evaluation Methodology

5.1 The Empirical Fallacy of LLM-as-a-Judge for Statutory RAG

Recent RAG literature relies heavily on synthetic LLM-as-a-judge frameworks such as RAGAS [29]. While the original RAGAS study demonstrated high human correlation on general-knowledge Wikipedia queries (WikiEval), its validity when deployed with lightweight on-premise models in public administration has remained unverified. We conducted an empirical calibration study (`eval/ragas_calib.py`) using local SLMs (Qwen3-4B and Qwen2.5-7B) to evaluate 92 administrative QA pairs against negative controls:

1. **Planted Hallucinations:** When an explicit fabrication (*"This procedure is completely free and completed within 1 hour at any municipal agency"*) was injected into legitimate answers, Qwen2.5-7B identified the hallucination in only 51% of cases (47/92), while Qwen3-4B caught only 30% (9/30).
2. **Inter-Judge Agreement:** Against an authoritative model baseline across 30 identical cases, Qwen2.5-7B achieved inter-annotator agreement of $\kappa = 0.02$ on Faithfulness and $\kappa = 0.20$ on Relevancy, falling drastically short of acceptable scientific reliability ($\kappa \geq 0.60$).

These findings demonstrate that **relying uncritically on LLM-as-a-judge scores creates a false sense of factual safety**. Consequently, Sys_3_4 establishes a strict architectural paradigm: administrative factuality is enforced via deterministic database anchoring and rule-based verification, rather than generative post-hoc grading.

5.2 Analysis of the Generalization Gap in Public Administration

A comparison of the controlled developmental suite (DEV, 98% Top-1) and the unseen blind suite (553 cases, 73% Top-1) highlights the primary challenges of real-world deployment:

- **Lexical Divergence:** Colloquial queries frequently employ dialectal verbs (*"xin giấy chứng sinh"*, *"nhập khẩu cho vợ"*) with low lexical overlap with codified procedure titles (*"Đăng ký khai sinh"*, *"Đăng ký thường trú"*).
- **Brother Procedure Ambiguity:** In civil status registration, closely related procedures (e.g., *Đăng ký kết hôn có yếu tố nước ngoài* vs. *Đăng ký kết hôn trong nước*) share over 80% of vocabulary. Without explicit clarification dialogue, single-turn search defaults to the domestic procedure, depressing Top-1 while maintaining high Top-3 recall (80%–87%).

5.3 Known System & Data Limitations

1. **Clarification Recall Bottleneck:** In the blind suite, only 47% of ambiguous inquiries successfully triggered clarification cards, with the remainder resolving to an assumed top candidate.
2. **National Portal Snapshot Omissions:** Standardized fees are published for only 467 of 1,350 procedures (34.6%). For the remaining 883 procedures, the portal records no explicit fee schedule; Sys_3_4 strictly outputs a refusal (*"Phí chưa được công bố chính thức"*) to avoid hallucination.
3. **Database String Parsing Anomalies:** Database audits revealed approximately 31 records where currency punctuation was truncated during scraper ingestion (e.g., `"20.000"` parsed as 20 VND instead of 20,000 VND). The strict engine implements regex-level currency sanitization to prevent misleading cost outputs.
4. **Data Provenance Verification:** To quantify the fidelity of the scraped gazetteer against the authoritative National Public Service Portal, we conducted a manual provenance audit over a random sample of 50 procedure records, cross-checking fees, processing times, legal basis, and decision numbers against the live portal pages. The audit yielded a source-faithful rate of 84.4% (211/250 fields; 95% CI: 79%–88%), rising to 93.5% (187/200; CI: 89%–96%) when restricted to the four core numerical/legal fields (fees, processing time, legal basis, decision number).

6 Conclusion & Future Work

This paper presented Sys_3_4, an enterprise context-aware legal artificial intelligence assistant designed for Vietnamese public administration. By integrating a deterministic **Strict Engine (System 3)** with an extensible **Friendly Engine (System 4)** and enforcing multi-turn *Clarify State Isolation*, the architecture resolves the twin hazards of hallucination and conversational drift. Extensive empirical evaluations across 1,352 standardized test cases demonstrate:

- **78% behavioral accuracy and 73% Top-1 retrieval** (MRR 0.759, nDCG@5 0.752) on the official 553-case blind evaluation suite.
- **85% behavioral accuracy and 78% Top-1** (MRR 0.824, nDCG@5 0.805) on the official HOLDOUT-4 benchmark.
- **91% general accuracy (29/32), 100% document QA citation accuracy, and zero web numerical hallucinations** on System 4.
- **100% PII redaction recall** on citizen identity numbers and telephone formats.
- **100% adversarial prompt injection resistance** across 11 distinct attack vectors.
- Sub-100 ms pure algorithmic retrieval latency on commodity CPU hardware.

Future work will focus on expanding text-to-SQL aggregation for administrative procedure listings, integrating regional dialectal query expansion, and implementing semantic vector caching (GPTCache) and continuous memory serving (vLLM) for provincial-scale deployment.

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